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Intracellular small molecule/ion storm from Ca-phytate nanoparticle synergizes neural stem cell therapy and immunomodulation in spinal cord injury

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ABSTRACT

Neural stem cells (NSCs) transplantation offers a promising therapeutic approach for spinal cord injury (SCI), addressing the challenge of neuron irreparability. However, the uncontrolled differentiation of transplanted NSCs and the inflammatory environment at the injury site limit the efficacy of cell-based treatments. To overcome these obstacles, simultaneously regulating NSCs differentiation and reshaping immune microenvironment is vital for enhancing the effectiveness of cell therapy. In this study, we proposed calcium phytate nanoparticles (Ca-PA NPs) as synergistic regulators capable of promoting neuronal differentiation and regulating the polarization of immune cells. This approach leverages the coordination properties of calcium ions and phytic acid, as well as the decomposition characteristics of calcium phytate. The nanoparticles can be endocytosed by NSCs or microglia, and decomposed in their lysosomes, forming the intracellular small molecule/ion storms. *In vitro*, NSCs endocytosing Ca-PA NPs predominantly differentiate into neurons, while microglia internalizing the nanoparticles transition from a pro-inflammatory M1 phenotype to a neuroprotective M2 phenotype. Importantly, the animal experiments proved that Ca-PA NPs significantly enhanced functional recovery in SCI mice by synergistically promoting neuronal differentiation of transplanted NSCs and reshaping immune-microenvironment. These findings open a new route to design nano-regulators for immune microenvironment involved stem cell therapy of SCI.

1. Introduction

Spinal cord injury (SCI) represents the impairment of the spinal cord brought about by trauma or pathological conditions, often leading to impairments in sensory, motor, and autonomic functions [1,2]. Typical causes include traffic accidents, falls, violent injuries, and sports-related incidents [3,4]. Globally, approximately 15–20 million people are affected by SCI, with around 900,000 new cases each year [5]. Beyond its seriously impact on individual health, SCI places a substantial burden on society and the economy.

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Following SCI, neuronal connections are disrupted, accompanied by the formation of pathological microenvironments characterized by inflammatory responses, glial scar development, and the release of inhibitory molecules, all of which further impede neural regeneration and functional recovery [6–9]. Due to the non-renewable nature of neurons, the transplantation of neural stem cells (NSCs), which possess the potential to differentiate into various types of neural cells, has emerged as a promising strategy for nerve regeneration. Notably, fetal brain-derived NSCs exhibit marked advantages over adult NSCs in transplantation due to their enhanced proliferation and differentiation capabilities [10–12]. Although stem cells transplantation has been clinically explored for SCI treatment, its therapeutic outcomes remain suboptimal, primarily due to the restrictive nature of the local microenvironment [13–15]. Microglia, as the resident immune cells of the central nervous system, are recruited to the injury site to modulate immune responses. However, the persistent activation of microglia triggers the excessive release of pro-inflammatory factors, impairing the survival and differentiation of transplanted NSCs and ultimately diminishing the effectiveness of cell-based therapies [16–22]. Therefore, a promising therapeutic strategy for SCI involves transplanting stem cells while simultaneously mitigating local inflammation to create a more favorable regenerative microenvironment. Targeted regulation of the fate of transplanted NSCs and microglia at the lesion site may be the key to achieving this goal. The fate of cells can be directed through exogenous physical and chemical methods, particularly chemical factors such as biological growth factors and small molecules [23–26]. In recent years, natural small-molecule compounds have gained significant attention for their ability to guide the differentiation of stem cells and immune cells by modulating specific signaling pathways and gene expression [27–30]. Building on the importance of optimizing the microenvironment for neural regeneration, natural small molecules with both neural differentiation and immune-modulatory properties hold promising potential for SCI treatment.

Phytic acid (PA), which is also called inositol hexaphosphate (IP6), is a compound that is extensively present in eukaryotic cells, exerting a vital regulatory function in the processes of cell proliferation and differentiation. It is mainly stored as a source of phosphorus in the seeds of plants like cereals, legumes, and nuts [31,32]. PA has been recognized by the Food and Drug Administration (FDA) as a nutraceutical. Recent studies have highlighted its potential in the prevention as well as treatment of neurodegenerative diseases due to its characteristics including metal ion chelation, antioxidant activity, anti-inflammatory function, and the ability to induce autophagy [33,34]. Notably, PA has been evidenced to restrain microglial activation and suppress the secretion of pro-inflammation cytokines [35]. Although direct evidence linking PA to the regulation of neural differentiation in stem cells is lacking, its modulation of cellular signaling pathways suggests potential effects on stem cell behaviors, including self-renewal and differentiation. Additionally, PA's anti-inflammatory properties may contribute to an environment that supports stem cell health and function, indirectly influencing differentiation outcomes. These attributes position PA as a promising candidate for stem cell therapy, where precise regulation of differentiation is essential. However, free PA faces significant limitations in the extracellular environment, where it readily chelates metal ions, forming diverse phytate complexes, and possibly loss its activity for regulation of stem cells. Moreover, PA relies on concentration-dependent diffusion into cells, which is inherently uncontrollable. To address these challenges, nano-delivery of PA *via* endocytosis has been explored, enhancing cellular uptake and bioavailability. This strategy has been validated in our previous studies, where carrier-free intracellular delivery of other small-molecule drugs proved effective [36–42]. Based on the structural characteristics of PA, which contains multiple phosphate groups, it can chelate with metal ions to form self-assemble nanoscale organic-metal complexes. Interestingly, metal ions, particularly calcium ions (Ca^{2+}), exert a crucial influence on regulating cell differentiation by operating as a secondary messenger inside numerous signaling pathways [43,44]. Therefore, the combination of PA and Ca^{2+} not only achieves the nanoization of PA molecules but also combines their biological activities, which is expected as a dual-functional approach to regulate NSCs differentiation and microglial activation.

In this study, self-assemble calcium phytate nanoparticles (Ca-PA NPs) were successfully synthesized *via* a chelation reaction. Once endocytosed by NSCs and microglia, the Ca-PA NPs decomposed in the mildly acidic environment of lysosomes, releasing both free PA and Ca^{2+}. The effects of Ca-PA NPs on NSC differentiation and microglial polarization were systematically evaluated. Additionally, the mechanisms underlying Ca-PA NPs-mediated neuronal differentiation and anti-inflammatory activity were investigated. Animal experiments were conducted to examine the impact of co-transplanting Ca-PA NPs with NSCs on neural regeneration in SCI mice, with a particular focus on neuronal differentiation and immune modulation.

2. Results and discussion

2.1. Characterization of Ca-PA NPs

The phosphate groups in PA readily chelate with metal ions, enabling the preparation of Ca-PA NPs by complexing PA with calcium ions. Fig. 1a illustrated the synthesis of Ca-PA NPs, where PA molecules reacted with calcium ions under mildly alkaline conditions (pH = 10), forming stable Ca-PA NPs through coordination between calcium ions and phosphate groups. The scanning electron microscopy (SEM) image of Ca-PA NPs revealed spherical nanoparticles approximately 50–100 nm in size (Fig. S1). The morphology of Ca-PA NPs was further examined using transmission electron microscopy (TEM), as shown in Fig. 1b, where multiple lattice structures were visible. Additionally, energy dispersive X-ray spectroscopy (EDX) analysis attested to the presence of vital elements in Ca-PA NPs (Fig. 1c). The elemental mapping of Ca-PA NPs showed the distribution of Ca, O, and P within the synthesized nanoparticles, suggesting an interaction between PA and calcium hydroxide.

The chemical structure of Ca-PA NPs was analyzed using Fourier transform infrared (FTIR) spectroscopy (Fig. 1d) and Raman spectroscopy (Fig. 1e), confirming the successful binding of calcium ions and phosphate groups. As shown in Fig. 1d, the characteristic peaks of PA were spotted at 500-900 cm^{-1} (*C*–*O*–P), 900-1200 cm^{-1} (PO_4), 1651 cm^{-1} (P═O), and 2900-3750 cm^{-1} (-OH) [45,46]. However, in the FTIR spectrum of Ca-PA, both the positions and intensities of these peaks shift. Specifically, Ca-PA exhibited stretching vibrations of PO_4 at 509 and 847 cm^{-1}, and bending vibrations of *C*–*O*–P at 995 and 116 cm^{-1}. These changes are attributed to the formation of Ca–*O*–P coordination bonds, indicating interactions between calcium ions and phosphate groups. Furthermore, the peak at 1651 cm^{-1} was significantly weakened, likely due to the interference of calcium ions with the vibrations of phosphate groups (P═O) [47]. Additionally, as shown in Fig. 1e, the characteristic peaks of *C*–*O*–P and PO_4 were located at 850 cm^{-1} and 1081 cm^{-1}, respectively [48]. Following the coordination binding between calcium ions and phosphate groups, a broad peak emerged at 117 cm^{-1}, indicating the formation of Ca–*O*–P. In addition, the differences between PA and Ca-PA in terms of crystalline state were compared through XRD (Fig. 1f). It can be seen that PA exhibited only a broad diffraction peak at 25°, whereas Ca-PA showed the distinct peaks at 17.93°, 29.39°, and 34.11°. This suggests that upon binding with calcium, PA transitions from an amorphous state to an organic phosphate with a crystalline structure, consistent with the TEM images. X-ray photoelectron spectroscopy (XPS) was utilized to further analyze the surface chemical status of elements in Ca-PA (Fig. 1g). The doublet peaks located at 347.5 eV ($2p_{3/2}$) and 351.1 eV ($2p_{1/2}$) were assigned to Ca 2p. The P 2p spectrum in Ca-PA showed two peaks located at 134.8 and 135.7 eV, which are corresponding to P $2p_{3/2}$ and P $2p_{1/2}$. The two distinct O 1s binding energy maxima, observed at 531.3 eV and 533.1

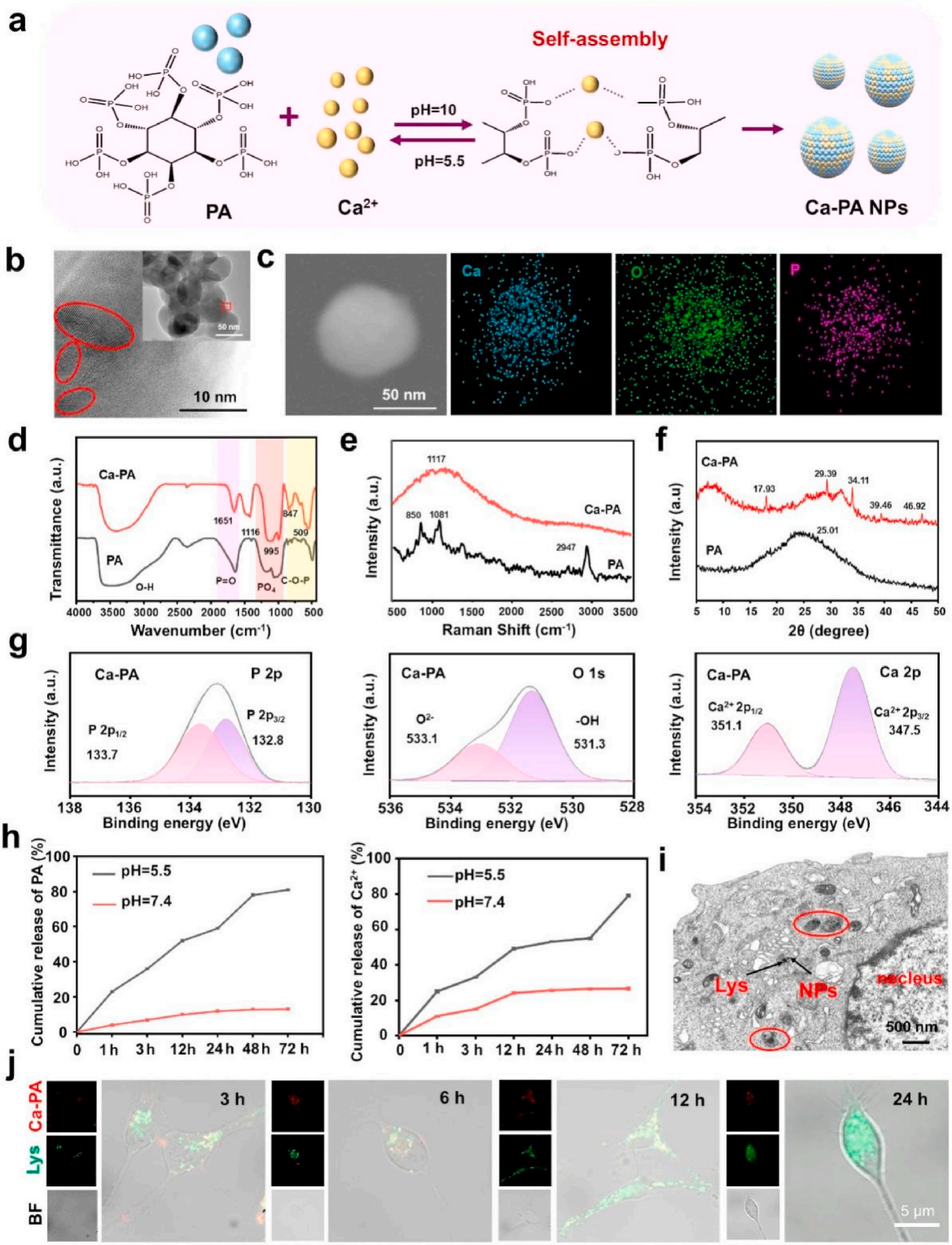


Fig. 1. Characterization of Ca-PA NPs. a) Schematic diagram of Ca-PA NPs synthesis. b) TEM images of Ca-PA NPs. c) EDX mapping images of Ca-PA NPs. Ca, O, and P elements are shown in blue, green, and purple, respectively d) FTIR spectra of PA and Ca-PA NPs. e) Raman spectra of PA and Ca-PA NPs. f) XRD patterns of PA and Ca-PA NPs. g) XPS spectra of the Ca-PA NPs for Ca 2p, P 2p and O 1s. h) Release of calcium ions and PA from Ca-PA NPs at pH 5.5 and pH 7.4. n = 3. i) TEM visualization of Ca-PA NP internalization in NSCs after 12 h. j) Fluorescence images of NSCs incubated with RhB-labeled Ca-PA NPs for 3 h, 6 h, 12 h, and 24 h. Lysosomes are stained with LysoTracker Green. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

eV, correspond to surface-bound hydroxyl groups (-OH) and O^{2-}, respectively. Additionally, the XPS spectra of $Ca(OH)_2$ and PA displayed the characteristic peaks of their respective surface elements (Fig. S2, Supporting Information). Comparing the peak position changes of Ca 2p, P 2p, and O 1s elements in three groups of samples suggests that calcium is bound to the phosphate groups of PA.

2.2. The neuronal differentiation effects of Ca-PA NPs on NSCs in vitro

2.2.1. Internalization and cytocompatibility of Ca-PA NPs with NSCs

Ca-PA NPs have been successfully prepared through the coordination binding of phosphate groups and calcium ions. However, it is necessary to verify whether Ca-PA NPs can be internalized by cells and decompose in weakly acidic lysosomes. Release of free PA molecules and ions plays a vital part in regulating the developmental outcome of NSCs with the involvement of Ca-PA NPs. Firstly, we evaluated the degradation characteristics of the Ca-PA NPs in the simulating lysosomal pH microenvironment. The release of Ca^{2+} and PA molecules was measured by employing inductively coupled plasma mass spectrometry (ICP-MS) and UV–visible spectrophotometry, respectively, with the results presented in Fig. 1h. At a pH of 7.4 (extracellular microenvironment), the Ca-PA NPs demonstrated a high degree of insolubility. Only minimal quantities of PA molecules and Ca^{2+} could be detected, indicating limited dissociation. In contrast, the release profiles of both PA molecules and Ca^{2+} exhibited a parallel upward trend over time. The cumulative release of PA molecules and Ca^{2+} achieved approximately 50 % by 12 h and further increased to around 80 % at 24 h. Additionally, SEM imaging of Ca-PA NPs dispersed within different pH systems revealed a significant reduction in particle volume and the emergence of a characteristic hemispherical morphology with a central indentation after being treated at pH 5.5 for 12 h (Fig. S3, Supporting Information). These results indicate that once the Ca-PA NPs enter the lysosome, they rapidly decompose and release their bioactive components within a short timeframe. Additionally, cellular internalization dynamics were characterized *via* fluorescence microscopy, which corroborated efficient uptake of Ca-PA NPs by NSCs. Following exposure of NSCs to rhodamine (RhB)-labeled nanoparticles (red) over 3, 6, 12, and 24 h, lysosomes were stained with LysoTracker Green (green). As visualized in Fig. 1j, nanoparticle distribution within lysosomes exhibited temporal dynamics: marked accumulation of red fluorescence was observed at 12 h, followed by attenuated signal intensity at 24 h, indicating that internalized NPs underwent progressive lysosomal sequestration followed by gradual degradation in the acidic microenvironment. TEM analysis further revealed Ca-PA NPs (≈100 nm diameter) localized within lysosomal vesicles after 12 h incubation (Fig. 1i). Collectively, these results confirm that Ca-PA NPs are efficiently internalized by NSCs and undergo lysosomal degradation in low-pH microenvironment.

Subsequently, we investigated the influence of different concentrations of Ca-PA NPs on the activity of NSCs through Live/dead staining and cell counting kit-8 (CCK-8) test. NSCs were cultured in proliferation media containing different concentrations of Ca-PA NPs for 1, 3, and 5 days, respectively. As demonstrated in Fig. S4a, exposure of NSCs to Ca-PA NPs at concentrations ≤200 μg/mL for 3 days resulted in no appreciable necrotic cells (red fluorescence). Notably, significant cell death became evident when the concentration was elevated to 400 μg/mL. Consistently, quantitative CCK-8 assay further revealed that cell viability remained comparable to the control group across all concentrations (25–200 μg/mL), whereas a marked reduction was observed at 400 μg/mL (Fig. S4b and S4c). In addition, the relative optical density (OD) values indicated that cell numbers in all groups gradually increased with prolonged culture, indicating good cell proliferation ability. However, differences were observed in the number of cells exposed to Ca-PA NPs groups. After 5 days of cultivation, no marked differences were observed between the 25 μg/mL and 100 μg/mL groups compared to the control group. In contrast, cell numbers in the 200 μg/mL and 400 μg/mL groups were notably less than those in the control group. Notably, the 50 μg/mL group showed a significantly higher cell number than the control. Collectively, these findings demonstrate a concentration-dependent effect of Ca-PA NPs on cell proliferation, with supraoptimal concentrations (≥200 μg/mL) eliciting cytotoxicity likely attributed to the abrupt intracellular release of calcium ions and PA. Based on these observations, concentrations of 25, 50, and 100 μg/mL were identified as cytocompatible with NSCs, thus selected for differentiation investigations.

2.2.2. Ca-PA NPs promote the differentiation of NSCs into neurons

It has been confirmed that Ca-PA NPs with appropriate concentration exhibit good cytocompatibility with NSCs. Quantitative real-time PCR (RT-qPCR) and western blotting (WB) analyses were subsequently conducted to evaluate the effects of Ca-PA NPs on the neuronal differentiation of NSCs. NSCs possess the ability to differentiate into three kinds of neural cells: neurons, astrocytes, and oligodendrocytes. β-tubulin III (Tuj1) and microtubule-associated protein 2 (MAP2) are neuron-specific markers, indicating immature and mature neurons, respectively. Glial fibrillary acidic protein (GFAP) serves as a specific marker for astrocytes, while oligodendrocyte transcription factor 2 (Olig2) and oligodendrocyte transcription factor4 (Olig4) are specific markers for oligodendrocytes. Here, we analyzed the marker genes and proteins of neural cells derived from NSCs induced by Ca-PA NPs for 5 and 10 days, respectively. As shown in Fig. 2a, the marker genes (*Tuj1* and *MAP2*) of neurons in Ca-PA NPs groups were significantly upregulated, while the marker genes (*GFAP* and *Olig2*) of glial cells were significantly downregulated. These results demonstrate that Ca-PA NPs effectively promote the differentiation of NSCs into neurons rather than glial cells, with the most significant effects observed at a concentration of 50 μg/mL. On the fifth day, the expression level of *Tuj1* was about 2.3-fold higher than that in the control group. By the tenth day, the expression level of *MAP2* had reached approximately 2.4-fold of that in the control group. These results imply that Ca-PA NPs might initially trigger the differentiation of NSCs into immature neurons and subsequently promote the maturation of these neurons.

To evaluate the effect of Ca-PA NPs enhancing the neuronal differentiation of NSCs, the marker proteins of differentiated neural cells were further analyzed. Immunofluorescence staining revealed neurons with Tuj1-positive or MAP2-positive (Tuj1+ or MAP2+) and astrocytes with GFAP-positive (GFAP+) displayed in red and green, respectively. As shown in Fig. 2b, after 5 days of differentiation culture, a higher number of neurons were observed in the Ca-PA NPs groups, whereas the control group showed a greater number of astrocytes. Moreover, quantitative analysis of cell numbers showed that Ca-PA NPs could induce the differentiation of NSCs into more neurons (Fig. 2c). Notably, Ca-PA NPs at a concentration of 50 μg/mL significantly promoted the neuronal differentiation of NSCs, with Tuj1+ cells for 60 % and MAP2+ cells for 50 %, compared to approximately 15 % in the control group. In addition, the immunofluorescence staining results at 10-days were consistent with those at 5-days (Fig. S5, Supporting Information). In the spontaneous differentiation state, NSCs differentiate into only a small number of neurons. However, the addition of Ca-PA NPs significantly enhances this process, leading to a substantially larger population of neurons. Bright field images, showing typical neuronal morphology, further indicated that the number of neurons differentiating from NSCs increased when cells were exposed to Ca-PA NPs (Fig. S6, Supporting Information). Additionally, WB assay was carried out to evaluate the proteins expression level of cells cultured with Ca-PA NPs. The marker proteins of neurons and glial cells were analyzed when NSCs were exposed to the Ca-PA NPs for 5 and 10 days. As shown in Fig. 2d, the levels of protein expression of Tuj1 and MAP2 in the Ca-PA NPs groups were significantly upregulated, while the expression levels of GFAP and Olig2 were notably downregulated compared with the control group. Especially, the 50 μg/mL group exhibited a more significant performance in enhancing the differentiation of NSCs to neurons compared to other concentration groups. The semi-quantitative analysis of protein expression showed that the expression level of MAP2 in the 50 μg/mL group was about 2-fold that of the control group. These WB results were consistent with the above RT-qPCR and immunofluorescence staining results. Interestingly, the phenotype of the differentiated neurons from NSCs was further characterized. Choline acetyltransferase (ChAT), glutamate decarboxylase 65 (GAD65), and tyrosine hydroxylase (TH) are specific markers for acetylcholine neurons, gamma aminobutyric neurons, and

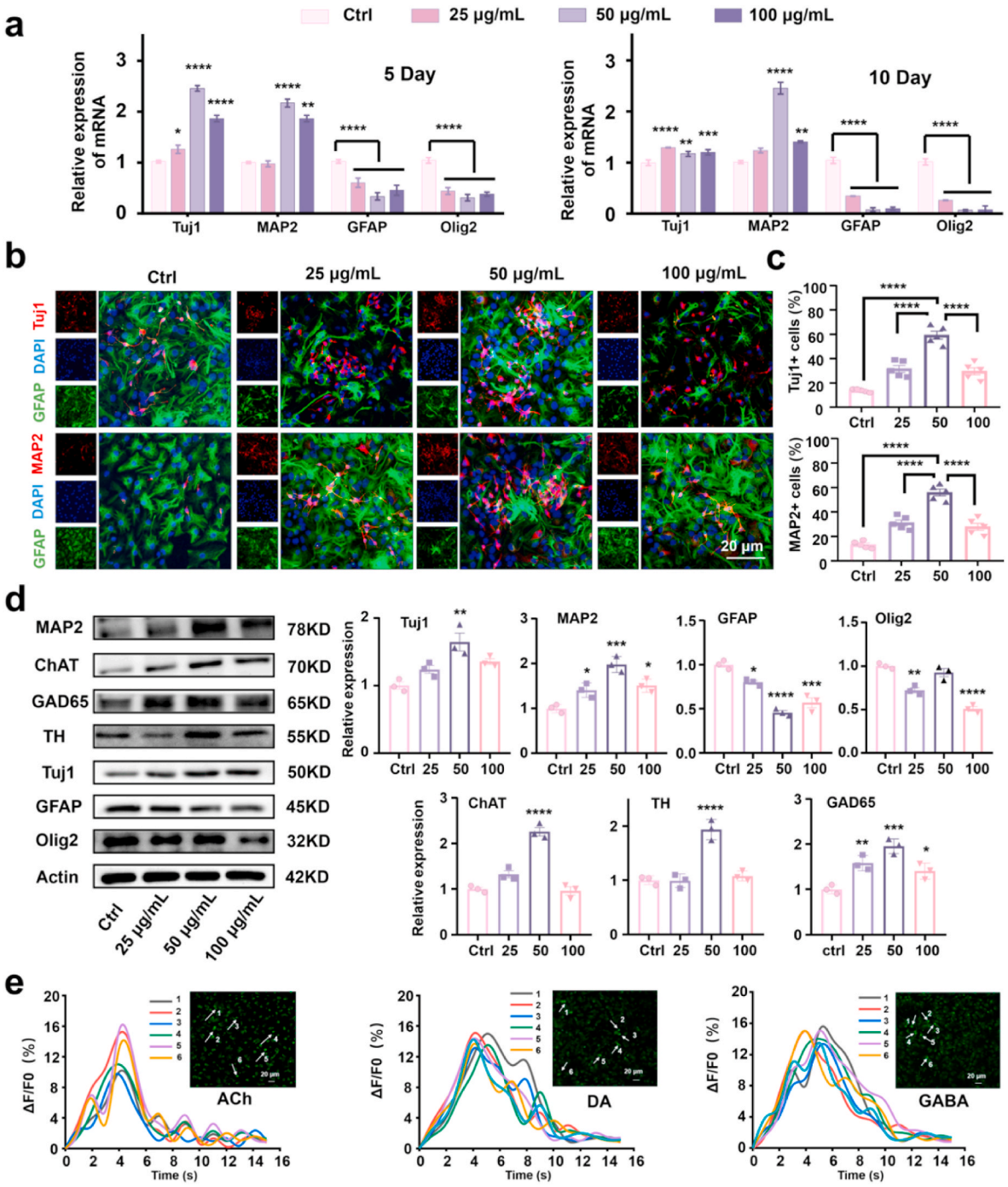


Fig. 2. The effect of Ca-PA NPs on neuronal differentiation of NSCs *in vitro*. a) RT-qPCR analysis of NSCs cultured with Ca-PA NPs at different concentrations for 5 and10 days. n = 6. b) Immunofluorescence staining images of Tuj1, MAP2 and GFAP in NSCs after culture with Ca-PA NPs at different concentrations for 5 days. Tuj1 and MAP2 were stained red, GFAP was stained green, and nuclei were stained blue by DAPI. c) Statistical analysis of the percentage of Tuj1 and MAP2 positive cells at day 5. n = 6. d) Western blotting and protein quantitative analysis for MAP2, ChAT, GAD65, TH, Tuj1, GFAP and Olig2 in NSCs cultured with Ca-PA NPs of different concentrations for 5 days. n = 3. e) Calcium spark analysis of the NSCs-derived neurons upon stimulation with acetylcholine (Ach), dopamine (DA) and γ-amino-butyric acid (GABA), respectively. $^{ns}p > 0.05$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$. Unless otherwise specified, all significance analyses represent comparisons with the control group. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

dopaminergic neurons, respectively. WB results revealed that cells treated with Ca-PA NPs for 5 days exhibited high protein expression levels of ChAT, GAD65, and TH. These findings indicate that Ca-PA NPs not only promote the differentiation of NSCs into neurons but also endow the differentiated neurons with potential functional plasticity. In addition, the protein expression analysis of cells induced by Ca-PA NPs for 10 days supported this conclusion (Fig. S7, Supporting Information). Therefore, it is confirmed at the gene and protein levels that Ca-PA NPs can efficiently guide NSCs to differentiate into more neurons.

In addition, the functional maturity of neurons differentiated from NSCs was evaluated by examining their physiological responses to neurotransmitters. After 5 days of culture with 50 μg/mL Ca-PA NPs, neurons were treated with Fluo-4AM, a Ca^{2+} indicator, and subsequently stimulated with acetylcholine (ACh), dopamine (DA), or

γ-aminobutyric acid (GABA). The fluorescence intensity of intracellular Ca^{2+} was recorded and time-lapse video imaging results were shown in Fig. 2e and Supplementary Movies S1-S6. Compared with the control group, cells in Ca-PA NPs group exhibited robust responses to neurotransmitter stimulation. Analysis of the fluorescence intensity of intracellular Ca^{2+} revealed that exposure to neurotransmitters triggered a rapid increase in fluorescence intensity, followed by a subsequent decrease in brightness (Fig. 2e). For instance, intracellular Ca^{2+} fluorescence peaked at 4 s after ACh stimulation and then rapidly diminished, demonstrating that neurotransmitters induce transmembrane ion flow, resulting in calcium influx. Furthermore, real-time Ca^{2+} fluorescence recordings clearly visualized intracellular calcium transients across multiple cells and their interconnections. A network of calcium signals was observed between cells, indicating that the differentiated neurons are capable of signal transmission. Collectively, these results demonstrate that neurons induced by Ca-PA NPs exhibit the

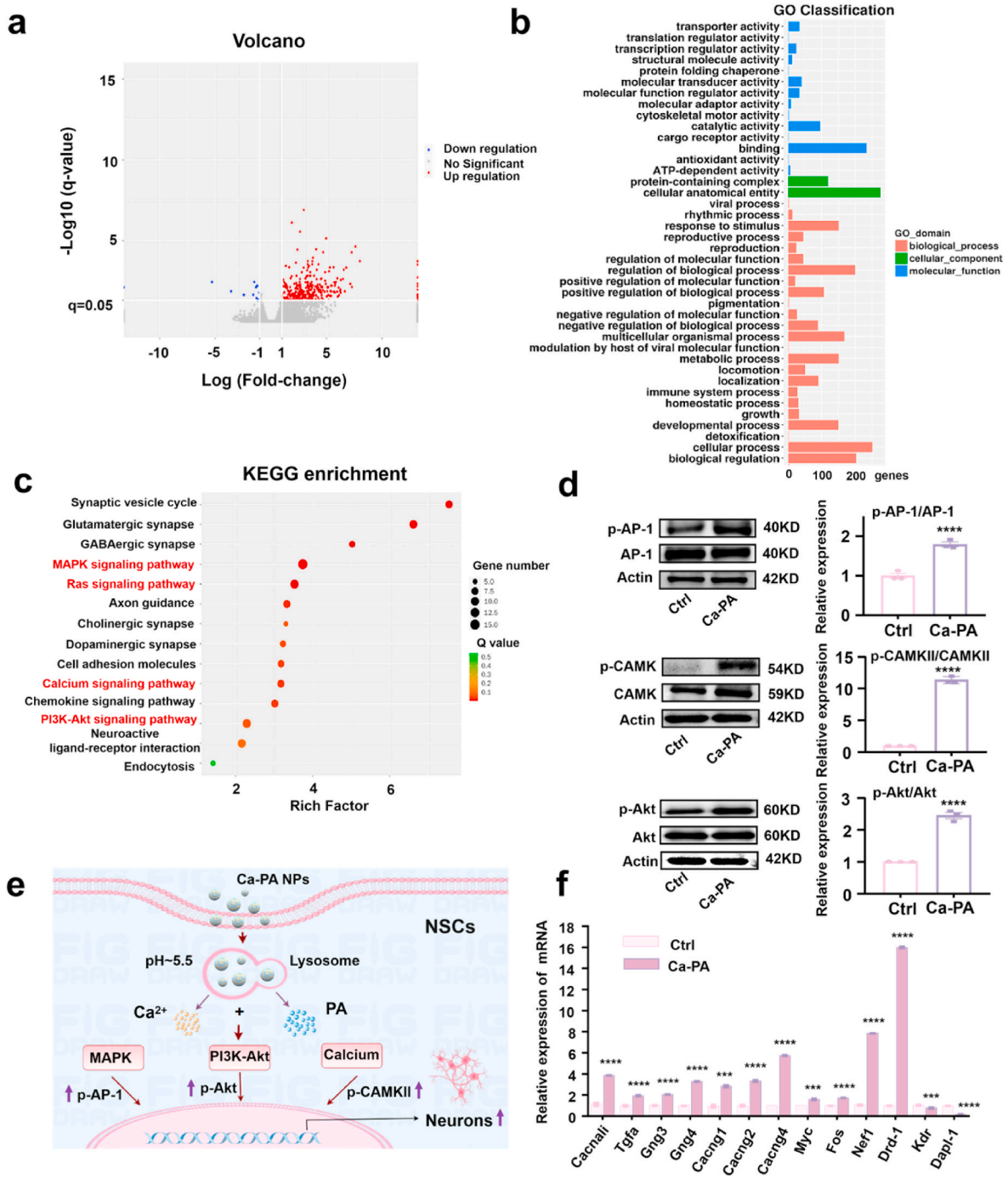


Fig. 3. Mechanism of Ca-PA NPs regulating the differentiation of NSCs into neurons. a) Volcano plot of the genes expressed differentially in cells treated with and without Ca-PA NPs for 5 days. The upregulated genes were shown in red, and the downregulated genes were shown in blue. b) The enriched GO terms of genes expressed differentially in (a). c) The enriched KEGG pathway terms of genes expressed differentially in (a). d) Western blotting analysis of AP-1, CAMKII and Akt proteins expression in key signaling pathways of KEGG pathway terms. e) Schematic diagram of mechanisms underlying Ca-PA NPs-induced the neuronal differentiation of NSCs. f) RT-qPCR analysis of representative genes expressed differently in (a). n = 3, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

functional characteristics of mature neurons.

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Multimedia component 2.

Multimedia component 3.

Multimedia component 4.

Multimedia component 5.

Multimedia component 6.

To further investigate the critical role of nanoscale properties in modulating neuronal differentiation of NSCs, the regulatory effects of Ca-PA NPs were compared with their free components. NSCs were incubated for 5 days with Ca-PA NPs (50 μg/mL), PA (41 μg/mL), Ca (9 μg/mL calcium chloride), or a combination of PA and Ca (41 μg/mL PA + 9 μg/mL calcium chloride), corresponding to the PA and Ca content in Ca-PA NPs. The protein expression levels of differentiated cells were assessed using WB and immunofluorescence analyses. As shown in Fig. S8, the Ca-PA NPs group demonstrated a significant upregulation of Tuj1 protein expression and a downregulation of GFAP and Olig2 protein levels, indicating that Ca-PA NPs are more effective than free components in promoting the differentiation of NSCs into neurons. Furthermore, immunofluorescence staining revealed a higher proportion of Tuj1+ and MAP2+ cells in the Ca-PA group compared to other groups, further substantiating these findings (Fig. S9). Although free Ca^{2+} and PA molecules can also enhance the neuronal differentiation of NSCs, their limited cellular internalization may constrain their effectiveness in regulating cellular outcomes. Therefore, Ca-PA NPs not only combine the biological functions of free PA and Ca^{2+} but also leverage nanoscale-mediated endocytosis to efficiently deliver bioactive components into cells, thereby triggering a series of cascade reactions within cells to regulate the differentiation of NSCs.

2.2.3. *Mechanism underlying the neuronal differentiation of NSCs regulated by Ca-PA NPs*

The upregulation of neuronal marker proteins and genes expression confirms that Ca-PA NPs can enhance the differentiation of NSCs into neurons. We further investigated the potential mechanism of Ca-PA-mediated neuronal differentiation of NSCs through mRNA sequencing analysis. The differentially expressed genes between the Ca-PA NPs group and the control group were analyzed, with the volcano plot presented in Fig. 3a. A total of 300 genes exhibited significantly differential expression patterns. Upregulated genes are shown in red, while downregulated genes are shown in blue. In addition, gene ontology (GO) analysis was performed to explore the biological functions and processes through which Ca-PA NPs regulate the fate of NSCs (Fig. 3b). Numerous significantly enriched GO categories are associated with biological processes and molecular functions. The biological processes primarily involve the regulation of biological process, metabolic process, and cellular process. Notably, key GO terms also highlight the developmental process and homeostatic process, which are closely linked to stem cell fate. The molecular functions encompass terms related to molecular sensor activity, catalytic activity, and binding.

Additionally, kyoto encyclopedia of genes and genomes (KEGG) results showed some crucial signaling pathways engaged in the control of NSCs differentiation by Ca-PA NPs. As shown in Fig. 3c, the main enriched signaling pathways were associated with the neuronal growth, including glutamatergic synapse, GABAergic synapse, cholinergic synapse, dopaminergic synapse and axon guidance. Notably, the typical enriched signaling pathways included the mitogenactivated protein kinase (MAPK) signaling pathway, rat sarcoma (Ras) signaling pathway, calcium signaling pathway, and phosphoinositide 3-kinase-protein kinase B (PI3K-Akt) signaling pathway. These signaling pathways are essential intracellular cascades that govern multiple cellular processes, including growth, survival, differentiation, and metabolism [49,50]. The MAPK and PI3K-Akt signaling pathways are particularly significant in stem cell differentiation. Studies have shown that the MAPK pathway modulates transcriptional regulator and cell fate determinants through the phosphorylation of its downstream effector [51,52]. As a key transcription factor, activator protein-1 (AP-1), which functions downstream of the MAPK pathway, regulates the expression of specific genes and thereby influences stem cell differentiation [53,54]. Similarly, the PI3K-Akt signaling pathway is critical in driving NSCs differentiation into neurons [55]. Akt promotes the expression of neuron-specific genes by activating various transcription factors and further supports neuron survival, synapse formation, and functional maturation [56–58]. To further demonstrate the critical roles of MAPK and PI3K-Akt signaling pathways in Ca-PA NPs-mediated differentiation of NSCs, the expression levels of AP-1 and Akt, including their total and phosphorylated forms, were analyzed independently. As shown in Fig. 3d, the relative expression levels of p-AP/AP and *p*-Akt/Akt were significantly upregulated during NSC differentiation induced by Ca-PA NPs. These findings suggest that PA and Ca^{2+} released from internalized Ca-PA NPs may contribute to the MAPK and PI3K-Akt signaling pathways, modulating the expression of differentiation-related genes through the activation of AP-1 and Akt phosphorylation. In addition, the KEGG enrichment results suggest that the calcium signaling pathway is also involved in the regulation of NSCs differentiation by Ca-PA NPs. Calcium/calmodulin-dependent protein kinase II (CaMKII) is a pivotal regulatory molecule in the calcium signaling pathway, with its function intricately linked to dynamic changes in calcium ion concentration [59,60]. As a key decoder of calcium signals, CaMKII translates calcium ion fluctuations into specific cellular responses, influencing processes such as cell differentiation and proliferation through the regulation of gene expression [61]. WB results showed that Ca-PA NPs promote the phosphorylation of CAMKII protein, leading to a significant upregulation of the *p*-CaMKII/CaMKII ratio. These findings collectively confirm that endocytosis-mediated delivery of PA and Ca^{2+} synergistically enhances the transformation of NSCs into neurons through the coupling of multiple signaling pathways, including the MAPK, PI3K-Akt, and calcium signaling pathways (Fig. 3e).

Furthermore, the key differentially expressed genes between the Ca-PA NPs group and the control group were validated using RT-qPCR analysis. As shown in Fig. 3f, the upregulation and downregulation of genes associated with signaling pathways, such as MAPK and PI3K-Akt, were consistent with the sequencing results. Calcium influx mediated by the calcium channel voltage-dependent alpha 1I (*Cacna1i*) activates calcium-dependent enzymes, such as CaMKII, which act as critical nodes linking calcium signaling to the MAPK pathway. The upregulation of *Cacna1i* enhances calcium influx, thereby activating calmodulin (CaM) and CaMKII [62,63]. Gamma subunits encoded by calcium voltage-gated channel auxiliary subunit genes (*Cacng1, Cacng2,* and *Cacng4*) primarily modulate the dynamic properties of voltage-gated calcium channels, influencing various biological processes by regulating intracellular calcium flow [64,65]. Additionally, the high expression of guanine nucleotide-binding protein genes (*GNG3* and *GNG4*) may enhance the activity of MAPK and PI3K-Akt pathways, including promoting Akt phosphorylation, thereby regulating cell differentiation [66,67]. Notably, dopamine receptor D1 (*Drd1*) serves as a key mediator of dopaminergic signaling, regulating neuronal excitability, synaptic plasticity, and learning and memory processes [68]. Neurofilament light polypeptide (*Nefl*) is highly expressed in neuronal axons, where it plays a critical role in providing structural support and facilitating signal transduction [69,70]. In contrast, genes such as kinase insert domain receptor (*Kdr*) and death-associated protein-like 1 (*Dapl1*), which are closely associated with cell apoptosis and differentiation, are inhibited, thereby blocking apoptosis-related pathways and promoting cell survival [71,72]. In summary, enrichment analysis of differentially expressed genes indicates that the mechanism by which Ca-PA NPs regulate the differentiation of NSCs into neurons may involve the integration of multiple key signaling pathways.

2.3. The anti-inflammatory effects of Ca-PA NPs on microglia in vitro

2.3.1. Internalization and cytocompatibility of Ca-PA NPs with BV2

Microglia exhibit both protective and detrimental effects in SCI, depending on their activation state and the dynamic changes within the injury microenvironment. Modulating microglial polarization and function is a promising therapeutic strategy to enhance recovery following injury. To further investigate the regulatory effect of Ca-PA NPs on the activation status of microglia, the phenotype of microglia after exposure to Ca-PA NPs was evaluated. Microglia, characterized by

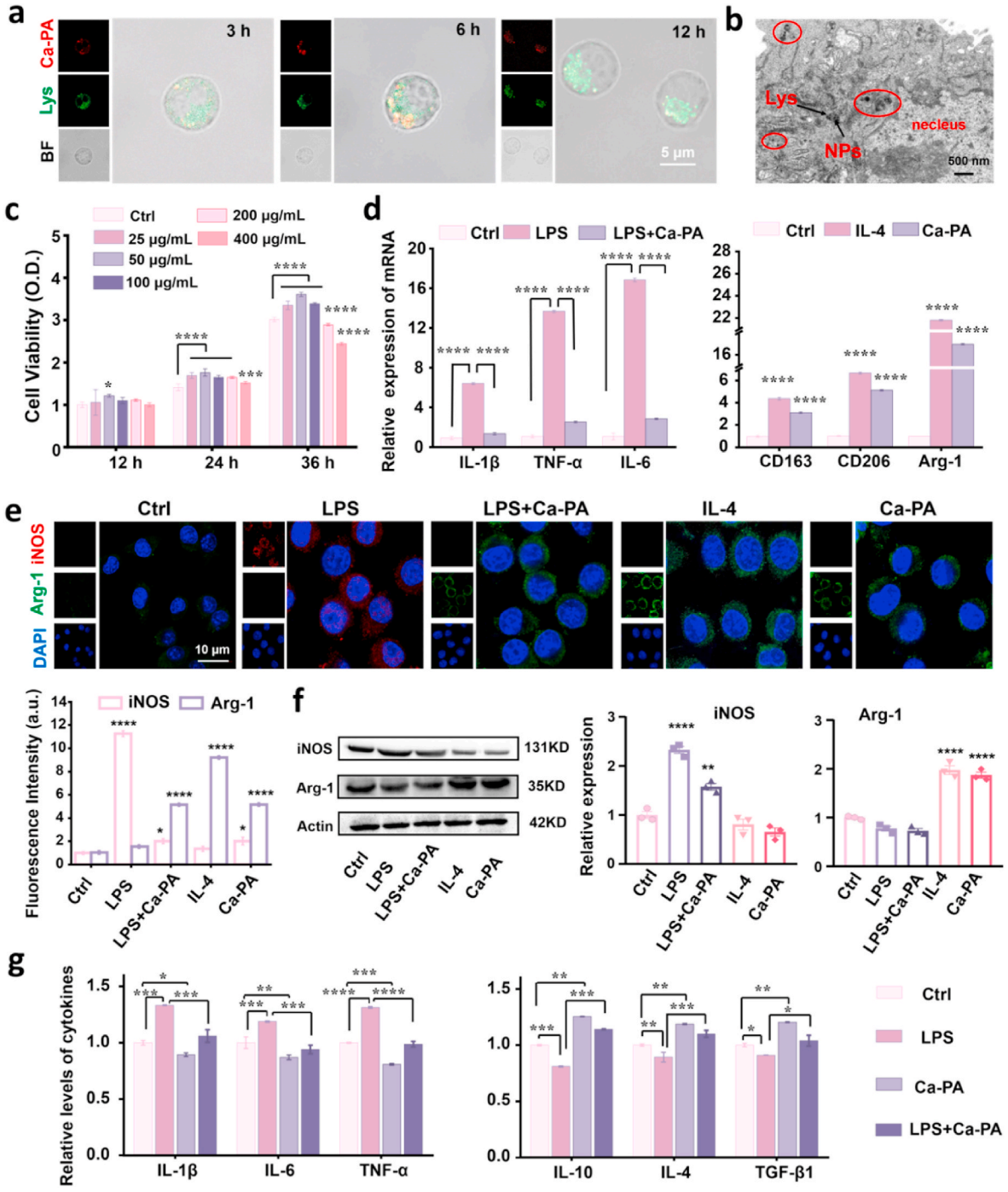


Fig. 4. The anti-inflammatory effects of Ca-PA NPs on microglia *in vitro*. a) Fluorescence images of BV2 cells exposed to RhB-labeled Ca-PA NPs for 3 h, 6 h and 12 h. Lysosomes were stained with LysoTracker Green. b) TEM images of BV2 cells following incubation with Ca-PA NPs for 6 h. c) Proliferation activity of BV2 cells cultured with Ca-PA NPs for 12, 24 and 36 h n = 6. d) RT-qPCR analysis of BV2 cells cultured under different conditions. The mRNA expression levels of *IL-1β*, *TNF-α* and *IL-6* in cells exposed to LPS and LPS + Ca-PA NPs; the mRNA expression levels of *CD163*, *CD206* and *Arg-1* in cells exposed to IL-4 and Ca-PA NPs. n = 6. e) Immunofluorescence staining and fluorescence intensity analysis of BV2 cells exposed to LPS, LPS + Ca-PA, IL-4 and Ca-PA NPs. iNOS and Arg-1 were stained red and green, respectively. n = 6. f) Western blotting and protein quantitative analysis of iNOS and Arg-1 in BV2 cells after incubation with LPS, LPS + Ca-PA, IL-4 and Ca-PA NPs. n = 3. g) ELISA analysis for pro-inflammatory cytokines (TNF-α, IL-1β and IL-6) and anti-inflammatory cytokines (IL-4, TGF-β1 and IL-10). n = 3. $^{ns}p > 0.05$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$. Unless otherwise specified, all significance analyses represent comparisons with the control group. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

specific markers, can be divided into pro-inflammatory M1 and anti-inflammatory M2 phenotypes. BV2 cells, which possess microglial characteristics, are extensively employed as a cell model for inflammation research. Prior to investigating cell phenotypes, we initially verified the internalization of Ca-PA NPs by BV2 cells. As shown in Fig. 4a, after 3 h of incubation, the nanoparticles had entered the cellular lysosomes, as manifested by the fluorescence co-localization of lysosomes and Ca-PA NPs. At 6 h, the fluorescence co-localization rate of nanoparticles and lysosomes within the cells peaked, suggesting that the internalization of Ca-PA NPs had likely reached its maximum extent. By 12 h, a decline in the fluorescence intensity of the nanoparticles was noted, presumably attributed to their degradation within the lysosomes. Moreover, TEM imaging of BV2 cells exposed to Ca-PA NPs for 12 h clearly demonstrated the presence of nanoparticles within lysosomes, as indicated by the red circles in Fig. 4b. Intriguingly, microglia internalize Ca-PA NPs more rapidly than NSCs, presumably due to their function in immune surveillance. In addition, we analyzed the effects of varying concentrations of Ca-PA NPs on the viability of microglia. As shown in Fig. 4c, BV2 cells exhibited good cell proliferation in the presence of Ca-PA NPs. Some concentration groups showed significantly higher cell numbers compared to the control group. The CCK-8 assay results indicate that Ca-PA NPs display excellent compatibility with BV2 cells, especially at concentrations below 400 μg/mL. This conclusion was further corroborated by the Live/dead staining results (Fig. S10, Supporting Information). In summary, Ca-PA NPs, which exhibit good cytocompatibility, can be effectively internalized into the lysosomes of microglia.

2.3.2. Ca-PA NPs regulate the M1/M2 phenotype of BV2

We have previously determined that Ca-PA NPs at a concentration of 50 μg/mL exert the most significant effect on regulating the neuronal differentiation NSCs. Moreover, this concentration also exhibits excellent compatibility with microglia. Consequently, Ca-PA NPs at a concentration of 50 μg/mL were subsequently utilized for investigating the M1/M2 phenotype of microglia. The expression levels of M1/M2 marker genes and proteins were analyzed by RT-qPCR and WB assays, respectively. Lipopolysaccharide (LPS) is a potent bacterial endotoxin commonly used in research to stimulate an immune response. In microglia, LPS triggers polarization towards a pro-inflammatory phenotype (M1). A cytokine known as interleukin-4 (IL-4) assumes a critical role in governing immune responses. When used to stimulate microglia, IL-4 promotes polarization towards an anti-inflammatory phenotype (M2). Here, M1 microglia activated by LPS and M2 microglia activated by IL-4 were used as positive control groups. The gene expression analysis for M1 markers (*TNF-α*, *IL-6*, and *IL-1β*) and M2 markers (*Arg-1*, *CD163*, and *CD206*) was shown in Fig. 4d. The gene expression levels of *TNF-α*, *IL-6*, and *IL-1 β* in cells significantly increased following LPS exposure. However, the addition of Ca-PA NPs notably reduced the expression of these inflammatory genes. These results suggest that Ca-PA NPs can inhibit the LPS-induced transformation of microglia into the M1 phenotype. Furthermore, when cells were exposed to IL-4 stimulation, the gene expression levels of *Arg-1*, *CD163*, and *CD206* significantly increased. Notably, Ca-PA NPs exhibited a similar effect to IL-4 in promoting microglia polarization towards the M2 phenotype. These results collectively suggest that Ca-PA NPs can inhibit the gene expression associated with the M1 phenotype while upregulating that of the M2 phenotype.

To further investigate the regulatory effect of Ca-PA NPs on microglia activation, we analyzed the expression of M1 and M2 phenotype marker proteins in microglia. Inducible nitric oxide synthase (iNOS) and Arginase-1 (Arg-1) are common marker proteins for the M1 and M2 phenotypes in microglia. iNOS is typically associated with the inflammatory response of M1 microglia, while Arg-1 is linked to the immune regulation and repair functions of M2 microglia. As shown in Fig. 4e, immunofluorescence images exhibited a significant increase in the number of iNOS-positive (iNOS+) cells in the LPS group, while the number of Arg-1-positive (Arg-1+) cells is significantly higher in the IL-4 group. In the presence of Ca-PA NPs, the quantity of iNOS + cells notably declined, while the quantity of Arg-1+ cells remarkably rose. The fluorescence intensity for iNOS and Arg-1 proteins was also quantified, which demonstrated that Ca-PA NPs inhibited the inflammation caused by LPS. In addition, the protein expression levels of iNOS and Arg-1 were evaluated by WB analysis (Fig. 4f). The protein expression level of iNOS in the LPS + Ca-PA group was significantly downregulated compared to the LPS group. Moreover, the protein expression level of Arg-1 in the Ca-PA group is comparable to that in the IL-4 group. These results further confirm the potential of Ca-PA NPs in inhibiting LPS-induced pro-inflammatory phenotype and enhancing anti-inflammatory phenotype. Furthermore, the regulatory effect of Ca-PA NPs on microglia was compared with that of free components (Fig. S11, Supporting Information). Following the addition of free Ca^{2+} and PA molecules, the proteins expression of iNOS and Arg-1 in LPS-induced BV2 cells showed no significant changes, indicating that the free components at equivalent concentration as Ca-PA NPs possess no anti-inflammatory effects. In the presence of 50 μg/mL Ca-PA NPs, the expression of the pro-inflammatory protein iNOS significantly increased, whereas the expression of the anti-inflammatory protein Arg-1 decreased. These findings suggest that endocytosis-mediated nano-delivery of PA and Ca^{2+} exhibits a superior cellular response, promoting the phenotypic transition of microglia from the M1 to the M2 state.

Microglia can transform into two distinct phenotypes, M1 and M2. The M1 phenotype is responsible for secreting pro-inflammatory cytokines, like TNF-α, IL-1β, and IL-6, and the M2 phenotype for secreting anti-inflammatory cytokines, such as IL-4, TGF-β1, and IL-10.These cytokines were further analyzed using ELISA to validate the regulatory effect of Ca-PA NPs on the M1/M2 phenotype of microglia (Fig. 4g). Quantitative analysis of cytokine expression revealed that Ca-PA NPs significantly inhibited the release of inflammatory cytokines induced by LPS, while enhancing the secretion of anti-inflammatory cytokines. Furthermore, compared with the control group, the Ca-PA group presented an elevation in anti-inflammatory cytokine production and a reduction in pro-inflammatory cytokine production. In summary, the results from gene, protein, and cytokine analyses demonstrate that Ca-PA NPs play an important role in regulating the activation state of microglia. Ca-PA NPs can inhibit excessive inflammatory activation of microglia and induce their progression from a pro-inflammatory phenotype to an anti-inflammatory phenotype, providing a suitable microenvironment to promote neural regeneration.

2.3.3. Mechanism underlying the M1/M2 polarization of BV2 regulated by Ca-PA NPs

To investigate the mechanisms by which Ca-PA NPs regulate microglial phenotypes, mRNA sequencing was performed to analyze the differentially expressed genes between LPS and LPS + Ca-PA NPs groups. As shown in Fig. 5a, the heatmap highlights the differentially expressed genes between two groups, identifying 133 upregulated genes (in red) and 122 downregulated genes (in blue). Additionally, GO analysis was conducted to enrich these differentially expressed genes, with the top 30 GO enrichment terms presented in Fig. 5b. Several significantly enriched GO terms are related to biological processes, molecular functions, and cellular components. Notably, many categories are related to inflammation, including positive regulation of inflammatory response, negative regulation of cytokine production, and immunoglobulin receptor binding. Furthermore, molecular functions include antioxidant-related categories such as antioxidant activity, oxygen transporter activity, and oxygen binding. These findings underscore the role of Ca-PA NPs in modulating microglial inflammatory responses and enhancing antioxidant activity. In addition, the related differentially expressed genes were further analyzed using KEGG enrichment to identify the main signaling pathways (Fig. 5c). These signaling pathways mainly involve pathways related to inflammatory regulation, especially the chemokine signaling pathway, janus kinase-

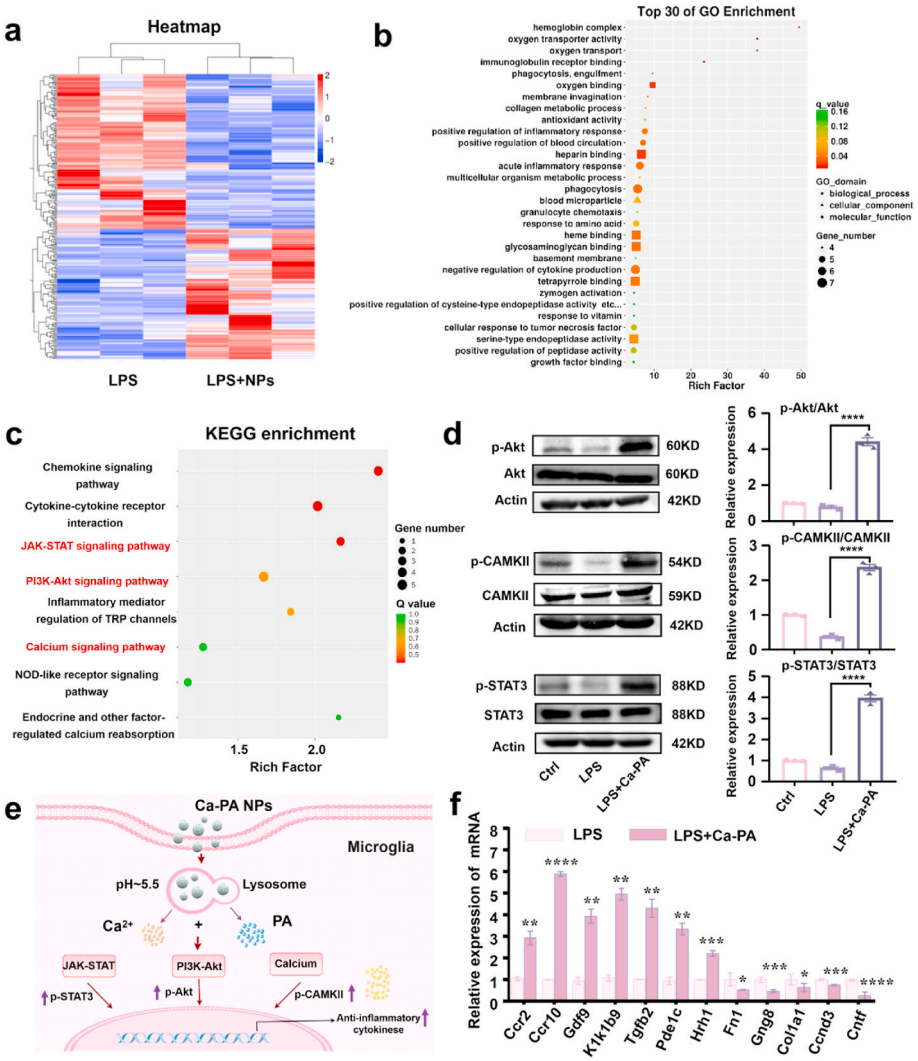


Fig. 5. Mechanism of Ca-PA NPs regulating the M1/M2 polarization of microglia. a) Heatmap of the genes expressed differentially in BV2 cells between treated with LPS and LPS + Ca-PA NPs. The upregulated genes were shown in red, and the downregulated genes were shown in blue. b) The enriched GO terms of genes expressed differentially in (a). c)The enriched KEGG pathway terms of genes expressed differentially in (a). d) Western blotting analysis of Akt, CAMKII and STAT proteins expression in key signaling pathways of KEGG pathway terms. e) Schematic diagram of mechanisms underlying Ca-PA NPs-induced the M2 type polarization of microglia. f) RT-qPCR analysis of representative genes expressed differently in (a). n = 3, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, $****p < 0.0001$. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

signal transducer and activator of transcription (JAK-STAT) signaling pathway, PI3K-Akt signaling pathway, calcium signaling pathway, and NOD-like receptor signaling pathway.

The JAK-STAT signaling pathway serves as a pivotal regulator of microglial polarization, directing their differentiation into M1 (pro-inflammatory) or M2 (anti-inflammatory) phenotypes through the activation of distinct STAT protein subtypes, including STAT1, STAT3, and STAT6 [73,74]. STAT1 acts as the key transcription factor for M1 polarization, driving the expression of pro-inflammatory genes. In contrast, STAT3 and STAT6 work to mitigate excessive M1 polarization while promoting the expression of M2-associated genes, thereby maintaining a balance between inflammation and tissue repair [75,76]. In addition, the PI3K-Akt signaling pathway assumes a pivotal regulatory part in microglial polarization, exerting inhibitory effects on M1 polarization while promoting M2 polarization [77]. Akt suppresses the expression of pro-inflammatory cytokines by inhibiting the nuclear factor-kappa B (NF-κB) signaling pathway and enhances the production of anti-inflammatory cytokines by activating mammalian target of rapamycin (mTOR) [78,79]. Previous studies have demonstrated that PA molecules influence the activation status of microglia through their antioxidant and anti-inflammatory properties [80]. Here, sequencing analysis further confirms the role of PA in regulating microglial phenotypes, particularly by modulating signaling pathways when delivered in nano-formulations. Interestingly, the enrichment pathway also highlights the role of the calcium signaling pathway in the regulation of M1/M2 polarization in microglia by Ca-PA NPs. The calcium signaling pathway regulates downstream effector molecules and pathways, often working in synergy with PI3K-Akt, JAK-STAT, and other signaling cascades to maintain the dynamic balance between M1 and M2 polarization in microglia [81,82]. Calcium ions released by nanoparticles are likely to activate downstream anti-inflammatory cascades by mediating the calcium signaling pathway. To further validate the impact of Ca-PA NPs on microglial polarization *via* these signaling pathways, the activation levels of key proteins were analyzed using WB assay. As shown in Fig. 5d, exposure to LPS significantly reduced the phosphorylation levels of STAT, Akt, and CaMKII, indicating that LPS promotes microglial polarization towards the M1 phenotype by suppressing the activation of these proteins. Conversely, the presence of Ca-PA NPs notably increased the p-STAT/STAT, *p*-Akt/Akt, and *p*-CaMKII/CaMKII ratios, suggesting that Ca-PA NPs induce the phosphorylation of these proteins and

promote microglial polarization towards the M2 phenotype. These results highlight the potential of Ca-PA NPs to regulate the M1/M2 polarization state of microglia by activating key signaling pathways, including JAK-STAT, PI3K-Akt and calcium signaling pathways (Fig. 5e).

To validate the sequencing results, we further performed experimental confirmation on several representative differentially expressed genes (Fig. 5f). Chemokine (*C*–C motif) receptor 2 (*Ccr2*) is a G protein-coupled receptor that plays a critical role in the immune system by mediating the recruitment of monocytes, macrophages, and other immune cells to sites of inflammation [83]. Chemokine (*C*–C motif) receptor 10 (*Ccr10*), another member of the chemokine receptor family, mediates immune cell trafficking and inflammation [84]. Additionally, genes closely associated with cytokine-cytokine receptor interactions, such as growth differentiation factor 9 (*Gdf9*), transforming growth factor beta 2 (*Tgfb2*), and kallikrein 1-related peptidase B9 (*Klk1b9*), exhibit high expression levels that facilitate the regulation of cytokine secretion, particularly anti-inflammatory cytokines [85,86]. Certain genes, particularly those associated with the JAK-STAT and PI3K-Akt pathways, play pivotal roles in regulating immune cell activation. Fibronectin 1 (*FN1*) binds to Toll-like receptors (TLRs), activating the NF-κB and MAPK signaling pathways, which enhances the secretion of

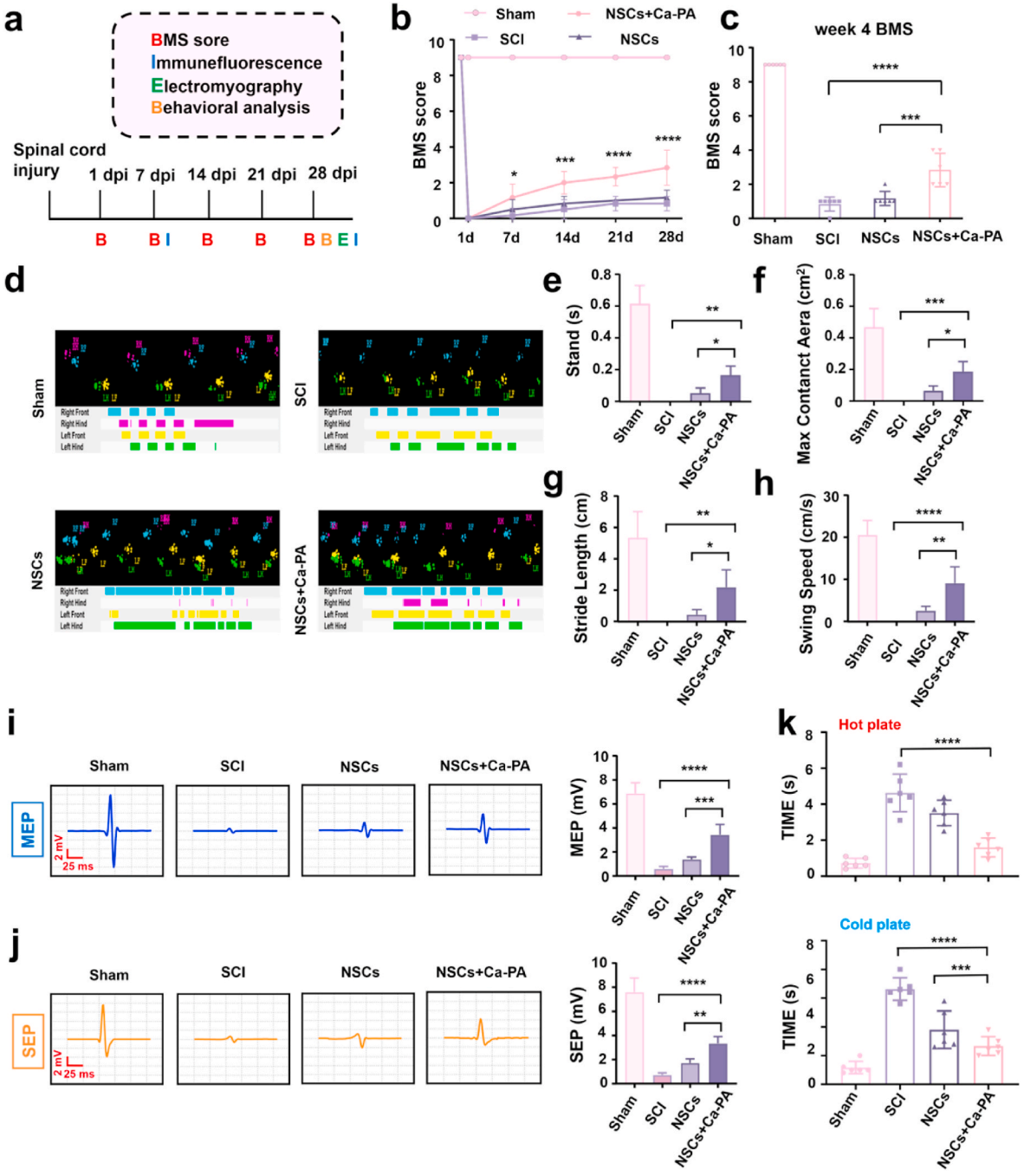


Fig. 6. The therapeutic effect of Ca-PA NPs on SCI mice. a) Schematic diagram of animal experiment on NSCs combined with Ca-PA therapy for SCI mice. b, c) BMS scores of SCI mice implanted with Ca-PA and NSCs. d-h) Gait analysis of SCI mice implanted with Ca-PA and NSCs. The parameter analysis included footprints, stand time, stride length, maximum contact area, and swing speed. i) MEP analysis of mice with spinal cord injury after treatment. j) SEP analysis of mice with spinal cord injury after treatment. k) Hot and cold plate tests and analysis for SCI mice. n = 6, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$.

pro-inflammatory cytokines [87]. Guanine nucleotide-binding protein gamma 8 (*GNG8*) facilitates inflammatory responses by promoting chemotaxis and immune cell activation [88,89]. Cyclin D3 (*CCND3*) and Ciliary Neurotrophic Factor (*CNTF*) engage the JAK/STAT and other signaling pathways, contributing to immune cell regulation and inflammation [90,91]. Importantly, the inhibition of these genes suggests that Ca-PA NPs can modulate microglial polarization by down-regulating key gene expression in specific signaling pathways. In summary, gene sequencing results demonstrate that Ca-PA NPs can not only guide the differentiation of NSCs but also regulate the activation phenotype of microglia at the gene and protein levels. These dual-functional nano-regulators hold significant promise for enhancing the microenvironment of SCI and accelerating neural repair by promoting neuronal differentiation and regulating immune responses.

2.4. Neural regeneration and immunomodulation in vivo

2.4.1. Functional recovery of SCI mice after treatment

To further investigate the role of Ca-PA NPs in neuronal differentiation and anti-inflammatory during SCI repair, a SCI hemi-transected model was employed to evaluate the *in vivo* performance of Ca-PA NPs. The schematic diagram of animal experiment on NSCs combined with Ca-PA therapy for SCI mice was shown in Fig. 6a. Firstly, the behavioral recovery of SCI mice after treatment was evaluated for 4 weeks. As illustrated in Fig. 6b and c, basso mouse scale (BMS) scores and gait analysis revealed motor function recovery in sham, SCI, NSCs, and NSCs + Ca-PA groups. Hindlimb locomotion across all experimental groups was assessed using the open-field locomotion test and post-injury BMS scores, which dropped to 0 in all animals on day 1 post-injury. Motor functional assessments indicated significant locomotion recovery in the first week following NSCs + Ca-PA treatment, reaching a peak in the fourth week. Animals in the SCI group and the NSCs transplantation-only group exhibited minor ankle movements, including spasticity, with a final mean BMS score below 2. Additionally, footprint analysis was employed to assess and quantify movement details at the final time points (Fig. 6d-h and Movies S7-S10). Mice were assessed using the Catwalk XT system, which recorded gait patterns as they traversed the illuminated pathway. Mice treated with transplanted NSCs + Ca-PA exhibited a significant increase in the standing time of the right hindlimb (Fig. 6e). Notably, the hindlimb swing duration increased 4 weeks after Ca-PA implantation (Fig. 6f). The step length of the right hindlimb, significantly reduced after spinal cord injury, was markedly increased following treatment compared to SCI controls (Fig. 6g). Furthermore, NSCs + Ca-PA transplantation markedly increased the maximum contact area of the right hindlimb compared to the NSCs-only and control groups (Fig. 6h). Additionally, neurological functional recovery was evaluated using motor evoked potentials (MEP) and sensory evoked potentials (SEP), which revealed increased amplitudes in the NSCs + Ca-PA group compared to the control and NSCs-only groups (Fig. 6i and j). The diminution or even complete loss of pain and temperature sensation following SCI remains a significant concern. To assess the restoration of nociceptive sensation, we performed a hot/cold plate experiment. The results revealed that the NSCs + Ca-PA group showed a shorter residence time, indicating a greater improvement in sensory function in the mice (Fig. 6k).

Following SCI, the loss of nervous function led to urinary retention in bladder, resulting in a significant decrease in the thickness of the bladder muscularis propria. H&E staining of the bladder revealed that the thickness of the bladder muscularis propria in the NSCs + Ca-PA group was significantly thicker than that in both the SCI and NSCs-only groups (Fig. S12, Supporting Information). Additionally, the NSCs + Ca-PA group exhibited significantly smaller bladder tissue thickness compared to the SCI and NSCs groups, suggesting that the addition of Ca-PA facilitated partial restoration of autonomic function. To verify the *in vivo* organ toxicity of the materials, the hearts, livers, spleens, lungs, and kidneys from all groups were collected at 4 weeks for H&E staining. The results showed no significant abnormalities between groups, confirming that Ca-PA demonstrated nontoxicity (Fig. S13, Supporting Information). These results indicate that Ca-PA co-transplanted with NSCs effectively mitigated motor and autonomic nervous functional deficits caused by SCI.

Supplementary data related to this article can be found online at https://doi.org/10.1016/j.biomaterials.2025.123708.

Multimedia component 7.

Multimedia component 8.

Multimedia component 9.

Multimedia component 10.

Multimedia component 11.

2.4.2. Neuronal differentiation and anti-inflammatory evaluation for SCI mice

Neuronal regeneration is considered crucial for recovery following SCI. Implantation of NSCs, along with the induction of differentiation and enhancement of axonal connectivity, are considered effective therapeutic strategies. To further investigate the mechanism by which Ca-PA promotes the differentiation of NSCs and neuronal regeneration *in vivo*, spinal cord tissues were histologically analyzed at 7 and 28 days post-injury. NSCs labeled with Dil cell membrane dye were added to the NSCs and NSCs + Ca-PA groups. After 7 days, the area of Tuj1-positive Dil (+) cells at the spinal cord injury site was significantly larger in the NSCs + Ca-PA group compared to the NSCs group. Additionally, the number of Tuj1-positive cells at the injury site was also markedly increased (Fig. 7a–c). These findings suggest that Ca-PA promotes the early-stage differentiation of transplanted NSCs into neuronal cells at the spinal cord injury site. At 28 days post-injury, NSCs + Ca-PA treatment continued to increase the number of Tuj1-positive cells, providing strong evidence that Ca-PA promotes neuronal regeneration (Fig. 7d and e). Spinal cord injury sites typically exhibit a high number of GFAP (+) astrocytes and the associated scar formation, which can severely hinder neuronal regeneration and the recovery of neurological function. Notably, the Ca-PA group showed significantly fewer GFAP-positive areas compared to the other groups, suggesting that Ca-PA effectively inhibits GFAP-positive astrocytes and their associated scar formation, thereby creating a favorable environment for neuronal regeneration (Fig. 7d and f).

The inflammatory response following SCI is a primary driver of secondary damage. Investigating strategies to inhibit this response and mitigate secondary injury is a critical area of research. The SCI group exhibited a higher number of CD68-positive cells at the injury site, indicating a robust inflammatory cascade activation and a pronounced inflammatory response (Fig. 8a and b). However, NSCs + Ca-PA treatment significantly reduced the number of CD68-positive cells at the injury site. Moreover, microglia play a central role in neuroinflammation, and their accumulation at the injury site is associated with exacerbated damage. Ca-PA treatment effectively reduced the number of Iba1-positive microglia at the injury site (Fig. 8a and c). To further examine the effects of Ca-PA on microglia, immunostaining was performed at the injury site. The results demonstrated a significant increase in Arg-1-positive cells and a marked decrease in iNOS-positive cells in the NSCs + Ca-PA group, opposite to the trend observed in the SCI group (Fig. 8d-f). These findings suggest that Ca-PA effectively suppresses the pro-inflammatory M1 phenotype of microglia in the injury center while promoting the anti-inflammatory M2 phenotype. Collectively, these findings indicate that Ca-PA can suppress the inflammatory response at the spinal cord injury site, thereby improving the local microenvironment and promoting neural regeneration.

3. Conclusions

In summary, calcium ions are bound to the phosphate group of PA through chelation reaction, thereby achieving the nanoization of PA molecules. The resulting Ca-PA NPs can be internalized by cells and

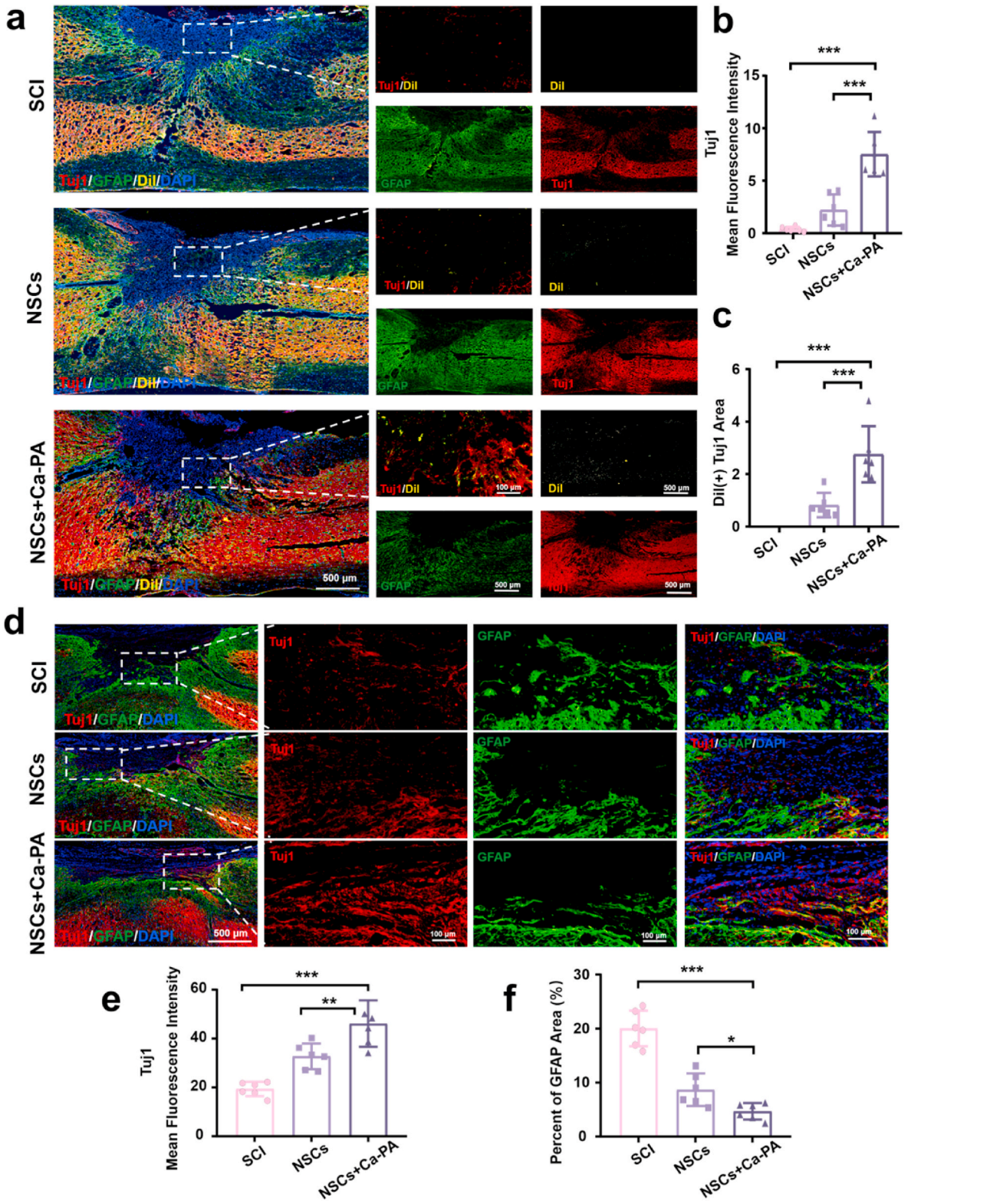


Fig. 7. Ca-PA NPs promote the neuronal differentiation of transplanted NSCs for neural regeneration in a semi-transverse SCI mouse model. a) Representative immunofluorescence staining images of Tuj1, GFAP, and Dil 7 days after spinal cord injury. b) Quantitative analysis of fluorescence area of Tuj1 in Dil (+). c) Quantitative analysis of fluorescence intensity of Tuj1. d) Representative fluorescence images of Tuj1 and GFAP 28 days after spinal cord injury. e, f) Quantitative analysis of Tuj1 fluorescence intensity and GFAP fluorescence area. n = 6, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

decomposed in lysosomal microenvironment to generate the intracellular small molecule/ion storm. On one hand, Ca-PA NPs enhance the differentiation of NSCs into mature neurons with a differentiation ratio of approximately 50 %, predominantly mediated through the MAPK, PI3K-Akt, and calcium signaling pathways. On the other hand, Ca-PA NPs inhibit M1 polarization of microglia while promoting M2 polarization, driven primarily by the JAK-STAT, PI3K-Akt, and calcium signaling pathways. Notably, the dual-functional Ca-PA NPs

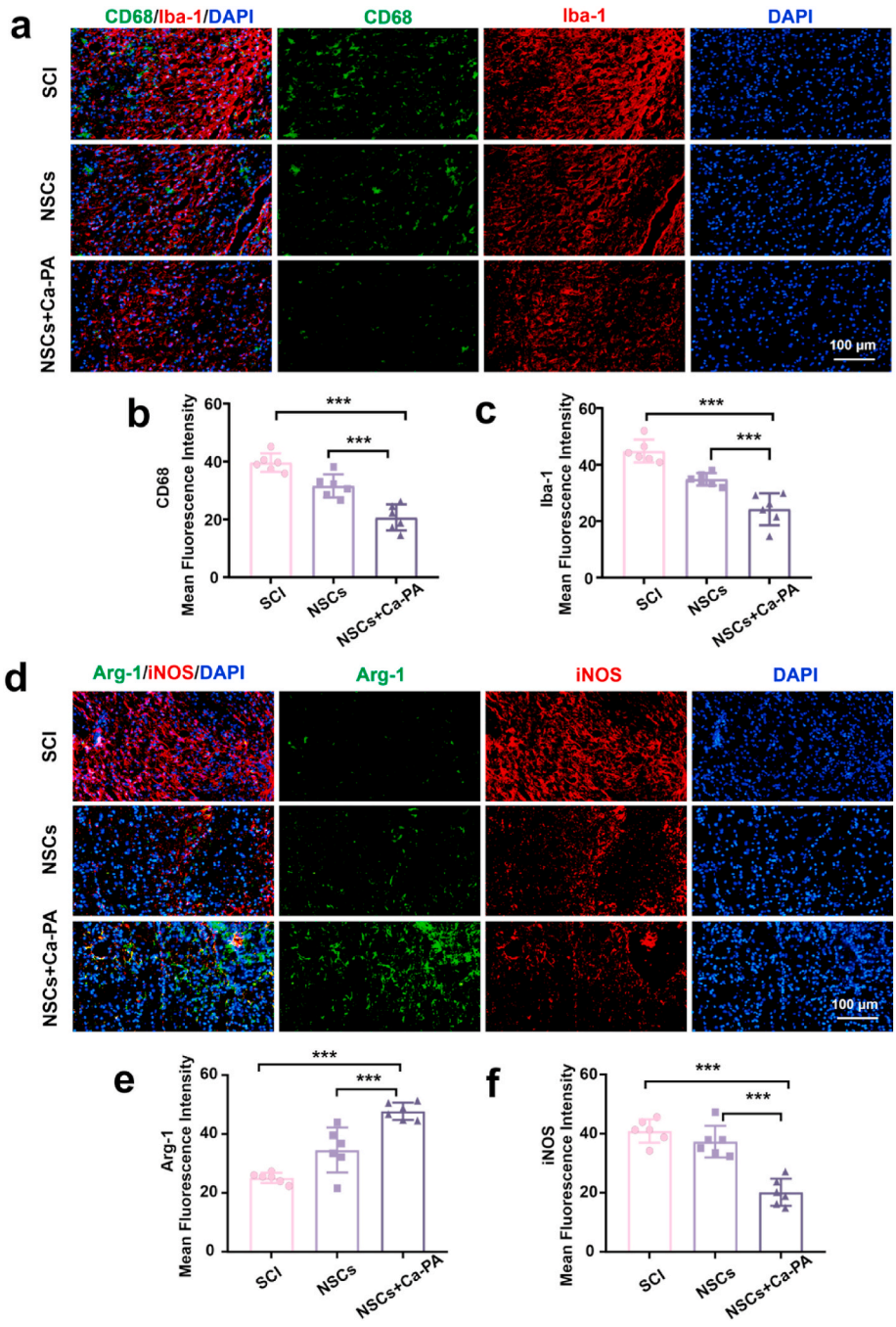


Fig. 8. Ca-PA inhibit inflammatory response by mediating M1 polarization of microglia after SCI. a) Representative fluorescence images of CD68 and Iba1, obtained 28 days post-SCI. b, c) Quantitative analysis of the mean fluorescence intensity of CD68, and the mean fluorescence intensity of Iba1. d) Representative fluorescence images of Arg-1 and iNOS, obtained 28 days post-SCI. e, f) Quantitative analysis of the mean fluorescence intensity of Arg-1, and the mean fluorescence intensity of iNOS. n = 6, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

significantly improve functional recovery in SCI mice, through their combined effects on neuronal differentiation and immune modulation.

4. Experimental section

4.1. Materials

Phytic acid (PA), calcium hydroxide, 3-aminopropyltriethoxysilane (APTES), and rhodamine B (RhB) were purchased from Maclin (China). Neurobasal medium, glutaMAX-1, fetal bovine serum (FBS), B27 supplement (B27), penicillin/streptomycin (P/S), high-glucose Dulbecco's Modified Eagle Medium (h-DMEM), StemPro™ Accutase™, and growth factors including epidermal growth factor (EGF) and basic fibroblast growth factor (bFGF) were sourced from Gibco (USA). The cell counting kit-8 (CCK-8) was obtained from Dojindo (Japan). Fluorescent probes such as LysoTracker (lysosome green), Dil (cell membrane yellow), Propidium iodide (PI), Calcein AM, and 4′, 6-diamidino-2-phenylindole (DAPI) were acquired from Beyotime Biotech (China). Lipopolysaccharide (LPS) and interleukin-4 (IL-4) were supplied by Solarbio (China). ELISA (enzyme linked immunosorbent assay) kits were purchased from Shanghai Enzyme linked Biotechnology Co., Ltd (China). RT-qPCR reagents were sourced from Takara (Japan). Antibodies including Tuj1, MAP2, GFAP, Olig2, TH, GAD65, ChAT, Arg-1, iNOS, AP-1, p-AP-1, Akt, *p*-Akt, STAT, p-STAT, CaMKII, and *p*-CaMKII

were obtained from Abcam (UK). Microglia (BV2 cells) were bought from ATCC (USA), and C57BL/6 mice were provided by Beijing Vital River Laboratory Animal Technology (Beijing, China).

4.2. Synthesis of Ca-PA NPs

A total of 7.41 g of calcium hydroxide was dissolved in 50 mL of deionized water under continuous stirring to ensure complete dissolution. Simultaneously, 13.2 g of PA solution (50 % w/w) was dissolved in another 50 mL of deionized water with stirring until fully homogenized. The calcium hydroxide solution was then gradually added to the vigorously stirred PA solution. Then 1 M NaOH solution was added dropwise to the mixture until the pH was adjusted to 10. The reaction mixture was stirred at 60 °C for 2 h to allow the reaction to proceed. The resulting product was collected *via* centrifugation and thoroughly washed with deionized water and ethanol. Finally, the material was freeze-dried for 24 h, yielding Ca-PA NPs.

4.3. Labeling Ca-PA NPs with RhB

The nanoparticles were functionalized with amino groups using APTES and subsequently conjugated with RhB. For functionalization, 300 mg of nanoparticles were dispersed in 10 mL of 90 % ethanol and reacted with 100 μL of APTES under vigorous stirring for 2 h. The modified nanoparticles were collected by centrifugation and washed with ethanol to remove excess APTES. Next, the amine-modified nanoparticles were suspended in 10 mL of ethanol containing 100 mg of RhB and stirred for 2 h. The RhB-labeled nanoparticles were isolated by centrifugation and washed with ethanol to remove unbound dye.

4.4. Characterization of Ca-PA NPs

The morphology of Ca-PA NPs was examined utilizing high-resolution transmission electron microscope (TEM, Gemini SEM 300) and scanning electron microscope (SEM, 8100, Hitachi, Japan). The elemental composition was determined through energy-dispersive X-ray spectroscopy (EDX, 8100, Hitachi, Japan) and X-ray photoelectron spectroscopy (XPS, Axis Supra, Kratos Japan). The crystallographic structure was analyzed with an X-ray diffractometer (XRD, ARL Equinox 3000, Thermo Fisher Scientific, USA). Furthermore, the chemical functional groups of Ca-PA NPs were identified using Fourier-transform infrared spectroscopy (FTIR, VERTEX 70, Bruker Germany) and Raman spectroscopy (LabRAM HR Evolution, Horiba France).

The release of calcium ions from Ca-PA NPs under physiologically relevant pH conditions was analyzed using inductively coupled plasma mass spectrometry (ICP-MS, PerkinElmer NexION 300D). Ca-PA NPs were dispersed in deionized water adjusted to pH 5.5 and 7.4, followed by dialysis against the respective pH solutions at 37 °C. Dialysate samples were collected at specific time intervals for ICP-MS analysis to monitor the calcium ion release profile. Additionally, UV–Vis spectrophotometry was employed to evaluate the release of PA molecules from the Ca-PA NPs in the dialysate samples.

4.5. Cell experiments

4.5.1. Isolation and culture of NSCs

NSCs were isolated from the hippocampal region of embryonic day 13–15 (E13-15) C57BL/6 mouse embryos (Beijing Vital River Laboratory Animal Technology). The NSCs were cultured in a proliferation medium containing neurobasal medium supplemented with 2 % B27, 1 % GlutaMAX-1, 20 ng/mL EGF and bFGF, and 1 % P/S. Once the neurospheres reached a diameter of 150 μm, they were collected and enzymatically dissociated with Accutase™ to generate a single-cell suspension for subculturing. For differentiation, the NSCs were cultured in a differentiation medium comprising neurobasal medium, 2 % B27, 1 % GlutaMAX-1, 2 % FBS, and 1 % P/S. NSCs from passages 2 to passages 5 (P2–P5) were utilized in this study. The experiments were prioritized for P2–P3 where possible, with P4–P5 used only for validation.

4.5.2. Culture of microglia

BV2 cells were used as a microglial model to study the anti-inflammatory effect of Ca-PA NPs. BV2 cell were cultured in a proliferation medium enriched with h-DMEM medium, 1 % P/S and 10 % FBS. BV2-M1 polarization was triggered by the proliferation medium with LPS at a concentration of 1 μg/mL, whereas BV2-M2 polarization was induced through exposure to the proliferation medium with IL-4 at a concentration of 100 ng/mL.

4.5.3. Cytocompatibility assay

NSCs or BV2 cells were seeded at a density of 5×10^4 cells/mL in 48-well plates and allowed to adhere overnight in proliferation medium. Then the cells were treated with fresh medium containing varying concentrations of Ca-PA NPs. Live/dead staining was performed after culturing NSCs and microglia for 3 days and 24 h, respectively. Briefly, 100 μL of Live/dead staining solution, comprising serum-free h-DMEM, 2 μM Calcein AM, and 4 μM PI, was added to each well. The samples were incubated at 37 °C for 30 min, then examined under a confocal laser scanning microscope (CLSM, LSM800, Carl Zeiss, German).

NSCs or BV2 cells were cultured in 96-well plates with or without Ca-PA NPs. Cell viability was assessed using the CCK-8 assay. A total of 100 μL of CCK-8 solution was added to each well, followed by incubation at 37 °C for 2 h. The optical density (OD) at 450 nm was accurately measured using a microplate reader (Spark, Tecan, Switzerland). To ensure the reliability of the results, the experiment was conducted in triplicate.

4.5.4. Cell uptake

NSCs or BV2 cells at a concentration of 5×10^4 cells/mL were seeded into 48-well plates and cultured for 12 h. The culture medium was then replaced with fresh medium containing 50 μg/mL RhB-labeled Ca-PA NPs. After incubation for 3, 6, 12, and 24 h, the cells were gently washed with PBS and treated with LysoTracker Green (1:500 dilution) for 30 min to visualize cellular lysosomes. Finally, fluorescence images were captured using a CLSM.

4.5.5. Immunofluorescence staining

After exposing to Ca-PA NPs for 5 and 10 days, the differentiated cells were subjected to immunofluorescence staining to identify the cell type. Meanwhile, the M1/M2 phenotype of microglia stimulated by Ca-PA NPs for 24 h was analyzed by immunofluorescence staining. Briefly, samples were washed with PBS and fixed in 4 % paraformaldehyde for 10 min. The cells were then permeabilized with 0.1 % Triton-X100 for 5 min to facilitate antibody penetration and blocked with 1 % bovine serum albumin (BSA) for 1 h to minimize non-specific binding. NSCs were incubated overnight at 4 °C with primary antibodies targeting Tuj1, MAP2, and GFAP, while BV2 cells were treated with primary antibodies against Arg-1 and iNOS under the same conditions. Following extensive PBS washes, the samples were incubated with Alexa Fluor 488/647-conjugated secondary antibodies. Before visualization using CLSM, the nuclei were stained with DAPI for 10 min. Fluorescence signals were quantitatively analyzed using ImageJ software, enabling precise measurement of immunofluorescence intensity.

4.5.6. ELISA

BV2 cells were exposed to LPS for 12 h, followed by treatment with Ca-PA NPs for 24 h. The cell culture medium was then collected for ELISA analysis. The samples were divided into four groups: control, LPS treatment, LPS + Ca-PA treatment, and Ca-PA treatment alone. The supernatant from the cell culture medium was carefully collected and centrifuged to remove cellular debris. The concentrations of IL-4, IL-10, TGF-β1, IL-1β, IL-6, and TNF-α, reflecting the cytokine profile of the

differentiated BV2 cells, were quantified using ELISA kits, following the manufacturers' protocols.

4.5.7. Calcium imaging

On the fifth day of Ca-PA NP treatment, NSCs were stained with 200 μL of 2 μM Flu-4AM working solution and incubated at 37 °C for 20 min. The cells were subsequently stimulated with distinct neurotransmitters, including dopamine (DA), acetylcholine (Ach), and gamma-aminobutyric acid (GABA). Changes in fluorescence intensity, indicative of calcium signaling, were recorded using a CLSM. The acquired images were analyzed with ImageJ software to quantify the cellular responses to neurotransmitter stimulation.

4.5.8. RT-qPCR

Total RNA was extracted from NSCs after 5-day and 10-day exposures to Ca-PA NPs at concentrations of 25 μg/mL, 50 μg/mL, and 100 μg/mL using Trizol reagent. Similarly, BV2 cells were treated with 50 μg/mL Ca-PA NPs for 24 h, and total RNA was isolated with the same reagent. Complementary DNA (cDNA) synthesis was performed using the polymerase chain reaction kit. Real-time quantitative PCR (RT-qPCR) was conducted using the LightCycler 96 system to analyze the expression of target genes, including β-actin, Tuj1, MAP2, GFAP, IL-1β, TNF-α, IL-6, Arg-1, CD163, and CD206 (primer sequences are provided in Supplementary Table S1 and S2). Each data was validated through triplicate experiments, with target gene expression levels normalized to β-actin as an internal control for consistency.

4.5.9. WB assay

NSCs were treated with Ca-PA NPs for 5 and 10 days, while BV2 cells were exposed to Ca-PA NPs for 24 h. Total protein was extracted using RIPA lysis buffer, and the protein concentration was measured at 562 nm with a multifunctional microplate reader (Synergy H1, BioTek, USA). Protein samples were analyzed *via* SDS-polyacrylamide gel electrophoresis (PAGE) and transferred onto a polyvinylidene fluoride (PVDF) membrane. After washing with TBST, the membrane was blocked with 10 % skim milk for 1 h at room temperature, followed by incubation with primary antibodies overnight at 4 °C. The membrane was then incubated with secondary antibodies for 1 h at room temperature. Finally, ECL reagent was applied for imaging, and ImageJ software was used for the semi-quantitative analysis of target protein expression.

4.5.10. mRNA sequencing

After treating NSCs and BV2 cells with 50 μg/mL Ca-PA NPs for 5 days and 24 h respectively, the total RNA of the cells was extracted for gene sequencing. Following quality control and quantification procedures, mRNA sequencing was performed on the BGISEQ500 platform at the Beijing Institute of Genomics. Three independent replicate samples were used for each group. Differentially expressed genes were identified and analyzed for functional and pathway enrichment using gene ontology (GO) and kyoto encyclopedia of genes and genomes (KEGG) pathways. Key differentially expressed genes and signaling pathways identified in the sequencing results were further validated through RT-qPCR and WB analysis.

4.6. Animal experiments

4.6.1. Mice spinal cord injury model

Initially, the animals received a mixture of Tribromoethanol (150 mg/kg) (i.p.) injection to induce deep anesthesia. A skin incision was made to expose the spinal column, followed by a laminectomy. A rostrocaudal incision along the spinal midline was made into the dura using a scalpel to expose the spinal cord. A 2-mm block of unilateral spinal cord tissue was carefully excised at T10. Subsequently, NSCs and Ca-PA NPs were added to the GelMA hydrogel and implanted into the defect. Each model mouse was injected with 10 μL of hydrogel containing NSCs (1×10^6) and Ca-PA NPs (50 μg/mL). A total of 24 C57BL/6 mice were used, all of which were female. The animals were randomly assigned into four groups: Sham group (n = 6), SCI group (n = 6), NSCs group (n = 6), and Ca-PA + NSCs group (n = 6). The paravertebral muscle layers were repositioned and sutured, and the skin was closed with staples. Following SCI, the bladder was emptied by hand twice daily. Intramuscular injection of cefuroxime sodium (6 mg/ml, dissolved in physiological saline; from Hongtu, Nanjing, China) was carried out to ward off urinary tract infections.

4.6.2. Gait analysis on catwalk

Mouse gait and locomotion were assessed using the CatWalk XT system and its accompanying software (version 10.6, Noldus, Wageningen, The Netherlands). Each mouse underwent three training sessions on the walkway prior to formal analysis. A digital video camera positioned beneath the glass walkway recorded the footprints of each mouse, which were subsequently analyzed using the CatWalk XT software. The analyzed parameters included stand time, stride length, maximum contact area, and swing speed.

4.6.3. BMS for locomotion

The mice were placed in a flat, enclosed field with a diameter of 100 cm. The locomotion assessment lasted for 4 min and involved two trained observers who evaluated hind limb joint movement, weight support, plantar stepping, and coordination. Scores ranged from 0 to 9, where 0 indicated complete paralysis and 9 denoted normal function. Mice were evaluated on day 1 post-injury and subsequently assessed weekly for 4 weeks.

4.6.4. Electrophysiology

Four weeks post-surgery, the mice were anesthetized with Tribromoethanol (150 mg/kg). Motor evoked potentials (MEP) and sensory evoked potentials (SEP) recordings were obtained using electrophysiological devices (BL-420L, TECHMAN). For each anesthetized mouse, the sciatic nerve and sensorimotor cortex were exposed. A plate stimulator and receptor were used, with the SEP stimulator applied to the sciatic nerve and the MEP setup being the reverse configuration. A constant stimulation of 10 mA current intensity was applied with a pulse width of 0.05 ms.

4.6.5. Mouse sensory evaluation

The hot and cold plate tests were conducted four weeks post-surgery. The hot and cold plates (BIO–CHP, Bioseb) were preheated to 55 °C or precooled to 1 °C. Once the target temperature was reached, the mice were placed on the plates, and the timing pedal was prepared. Timing began when the mouse was placed on the plate and ended upon observation of leg twitching or beating. An interval was maintained between the hot and cold tests to prevent overstimulation in the mice.

4.6.6. Immunohistochemistry

Mice were deeply anesthetized and subsequently perfused with precooled PBS followed by 4 % paraformaldehyde. The spinal cord was then fixed, dehydrated, embedded, and sectioned. The sections were washed three times with PBS, followed by incubation with a TBST solution containing 5 % normal goat plasma, 0.1 % BSA, and 0.25 % Triton X-100. Subsequently, primary antibodies, diluted in the sealing buffer, were applied to the sections and incubated at 4 °C overnight. The primary antibodies used in the experiments were Rabbit anti-beta III Tubulin (Tuj1) (1:500, Abcam, ab215037), Chicken anti-GFAP (1:500, Abcam, ab4674), Mouse anti-CD68 (1:50, Abcam, ab955), Rabbit anti-Iba1 (1:400, Abcam, ab178846), Mouse anti-iNOS (1:500, Abcam, ab210823), and Rabbit anti-Liver Arginase (Arg-1) (1:500, Abcam, ab96183). The sections were washed the following day and incubated with Alexa Fluor 488 and Alexa Fluor 647 (1:500, Invitrogen)-conjugated secondary antibodies for 2 h at room temperature. After washing, the sections were covered and imaged using both an inverted

fluorescence microscope and a confocal microscope. The fluorescence intensity of the sections was analyzed and quantified using ImageJ software.

4.6.7. H&E staining

To evaluate the biocompatibility of the nanocomplexes and assess the recovery of autonomic function, paraffin-embedded sections of the heart, liver, spleen, lungs, kidneys, and bladder were stained with hematoxylin and eosin (H&E), and organ changes were observed using light microscopy.

4.7. Statistical analysis

The data were expressed as mean ± standard deviation (SD) and analyzed by Origin Pro 2021 and GraphPad Prism software. Statistical analysis was conducted with one-way ANOVA. Statistical significance was defined as differences at $p \leq 0.05$, while "ns" denoted no significant difference between the two groups.

CRediT authorship contribution statement

Jiapei Shi: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Yanbin Feng:** Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Shuo Zhang:** Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Wenfeng Shao:** Methodology, Formal analysis, Data curation. **Na Ren:** Writing – review & editing, Supervision, Project administration. **Jingang Wang:** Writing – review & editing, Supervision, Project administration. **Chunhui Sun:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Zenan Wang:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Hengxing Zhou:** Writing – review & editing, Supervision, Project administration. **Hong Liu:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Shuping Wang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biomaterials.2025.123708.

Data availability

Data will be made available on request.

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